Lena Einramhof

MSc Student in Astronomy - University of Vienna
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EDUCATION

Oct 2025 – present (expected graduation: Aug 2027)

MSc in Astronomy

University of Vienna - Vienna, Austria

* Current average of 1.0
* Merit Scholarship (Leistungsstipendium) 2025/26

Oct 2022 – Aug 2025

BSc in Astronomy

University of Vienna - Vienna, Austria

* Graduated with average of 1.65 (Astronomy courses: 1.32, Programming courses: 1.0)
* Merit Scholarship (Leistungsstipendium) 2024/25
* 1st place, Coding Competition - BSc programming course

Feb 2016 – Jun 2022

Secondary School (Matura)

GRG Draschestraße - Vienna, Austria

* Graduated with average of 1.0
* Bilingual English program

RESEARCH EXPERIENCE

Jul 2026 – present

Funded Summer Research Internship & Master Project: Kinetic Sunyaev-Zel'dovich Effect & Velocity Reconstructions of BGS Galaxies

Supervisors of the Internship: Prof. Raul Angulo (DIPC) & Dr. Lurdes Ondaro-Mallea (Cambridge) - Computational Cosmology Group, DIPC, Donostia/San Sebastián, Spain

* Investigated stacked kSZ measurements of circumgalactic gas profiles of BGS tracers using non-linearly reconstructed velocities
* Performed linear velocity reconstructions and processed large-scale simulation data
* Presented progress in weekly meetings, collaborated with international researchers
* Started as an internship at DIPC, continues as my Master thesis in Vienna
* Funded by the Erasmus+ internship grant

Dec 2025 – Jun 2026

Paid Research Project: Galaxy Bias in the TNG300 Simulation

Supervisor: Prof. Oliver Hahn - Data Science in Astrophysics & Cosmology Group, University of Vienna

* Investigated spatial clustering, galaxy bias and secondary halo properties using large-scale simulation data
* Executed HPC workloads on the Vienna Scientific Cluster (VSC) using SLURM scheduling
* Developed, version-controlled and documented analysis code collaboratively
* Paper in preparation: Gaussian Galaxy Bias in the TNG300 Simulation

Jan – Aug 2025

Bachelor Project: Accurate Predictions of Density Field Statistics with Fast Simulations

Supervisor: Prof. Oliver Hahn - Data Science in Astrophysics & Cosmology Group, University of Vienna

* Systematic parameter study of Lagrangian perturbation theory, starting redshift and time-stepping using the DISCO-DJ framework
* Quantified accuracy–cost trade-offs via power spectrum, cross-correlation and higher-order statistics
* Preliminary results further developed in List et al. (2026)

PUBLICATIONS

Gaussian Galaxy Bias in the TNG300 Simulation
Lena Einramhof & Raffael Jonas (equal contribution), Oliver Hahn, Jens Stücker - in preparation

TEACHING & OUTREACH

Oct 2026 – Jun 2027
FIT (Fem* in Tech) Ambassador
FIT sprungbrett - Vienna, Austria
* Delivered workshops in schools for girls aged 15–18 to encourage women in STEM
* Addressed gender bias and the gender gap in tech, shared personal experiences of studying in a male-dominated field
* ~ 1 workshop per month

Sep 2025 – Feb 2026
Student Assistant (Tutor)
University of Vienna - Courses: Observational Lab (Spectroscopy) & Interstellar Medium and Star Formation
* Tutored students in Python-based spectroscopic data analysis: calibration, line fitting and classification
* Supported students with no prior programming experience, led weekly tutorials and graded coursework
* 4 h/week

Oct 2024 – Jun 2025
G:URL*s Coding Club
University of Vienna - Workshops and events for FLINTA* people in tech
* Participated in events including a C++ game development talk and an R bootcamp

OTHER EXPERIENCE

May 2024 – present
Student Job
Imperial Crypt (Kaisergruft) - Vienna, Austria
* Customer service, ticket sales and answering visitor questions about the crypt and its history
* 10 h/week (Jul–Aug 2024 & Feb 2025: 25 h/week)

Aug 2019
Internship
Kremsmueller Industrieanlagenbau - Schwechat, Austria
Support in the purchasing and sales department
* 38 h/week

SKILLS

Programming
Python (advanced) - NumPy, Astropy, Matplotlib, SciPy
C++ (intermediate)
Java (intermediate)
R (basics)

Tools & Methods
HPC & large-scale data analysis - VSC, SLURM scheduling, batch scripting
Simulations & Post-processing – IllustrisTNG data, HDF5, DISCO-DJ, ROCKSTAR halo finder
Open-Source Software Development – Git, version control, testing (pytest), dependency management
Astronomical Software - TOPCAT, SpectraWiz
Machine Learning - PyTorch, TensorFlow
LaTeX

LANGUAGES

German - Native **English** - Fluent (C1) **Spanish** - Intermediate (B2)